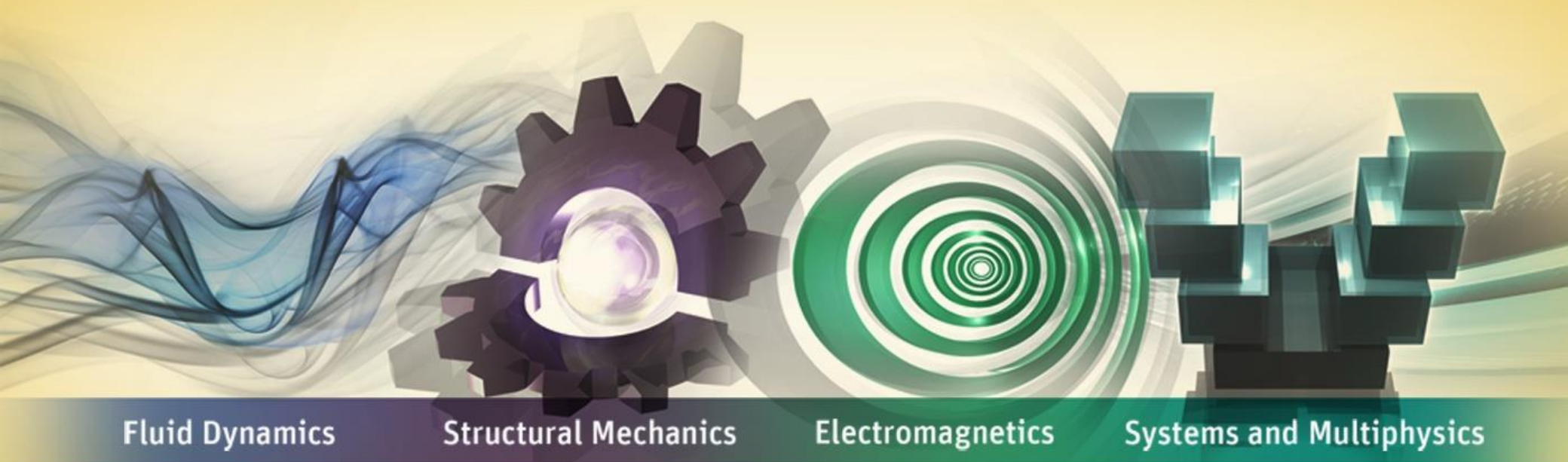


Lecture 1: Overview



ANSYS Mechanical

1.1 Fundamentals on linear elasticity

Why Study Materials?

- Selecting materials is part of the design process.
- The choice of materials is driven by several factors such as
 1. Cost
 2. Manufacturability
 3. Performance
 4. Availability
- A design engineer must consider all the above in picking the correct material.



Why Study Materials?

- Performance may include
 1. Mass-to-weight ratio
 2. Thermal and electrical conductivity
 3. Mechanical strength and behavior
 4. Corrosion and wear resistance
- All the above depend on the material's behavior.

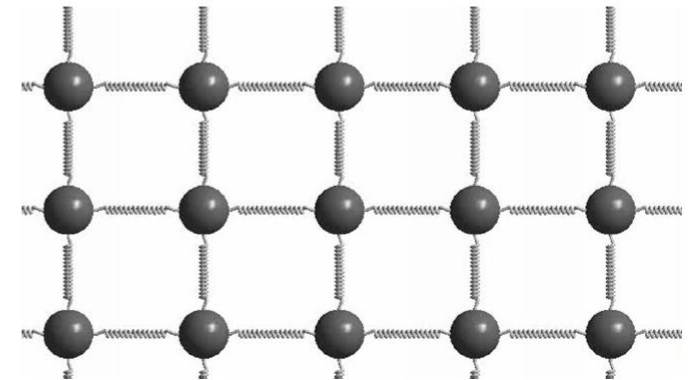


One must understand material behavior to pick the right material.



Material Behavior

- Material behavior is simply how a material responds to a physical environment.
- This can be structural, chemical, thermal, electrical or magnetic.
- All properties such as mass density, elastic modulus, thermal conductivity, etc., are considered as material properties.
- Each material has a unique value for its material properties.
- The difference in properties between materials is due to the difference in the microstructure and the molecular structure.



Material Behavior

- For instance, rubber offers smaller resistance to deformation compared to structural steel.
- So, rubber must stretch longer compared to structural steel before it can support the same dead weight.
- Quantifying this behavior in a mathematical equation is the essence of a material model.



Material Behavior

- Most engineering objects are assemblies of parts made of different materials.
- Each part has its own function and is made of a specific, relevant material.



Choosing Material for Chassis and Tires



Material criteria:

- Support the whole structure of the car
- Protect the car during a crash
- Material chosen:

- Structural steel
- Rubber



Easy choice!



Material criteria:

- Deform and provide traction
- Absorb small impacts and reduce the bumpiness
- Material chosen:

- Structural steel
- Rubber



Choosing Material for an Engine Block

Material Criteria:

- Withstand harsh thermal and mechanical environment
- Remain strong and durable in hot and corrosive environments
- Material chosen:
 - Cast iron
 - Hard steel



Not an easy choice!



Need for a Model

- Therefore we need a constitutive model!
- It'll help us both characterize and quantify material properties.
- Such a model helps engineers make informed decision on choosing materials.

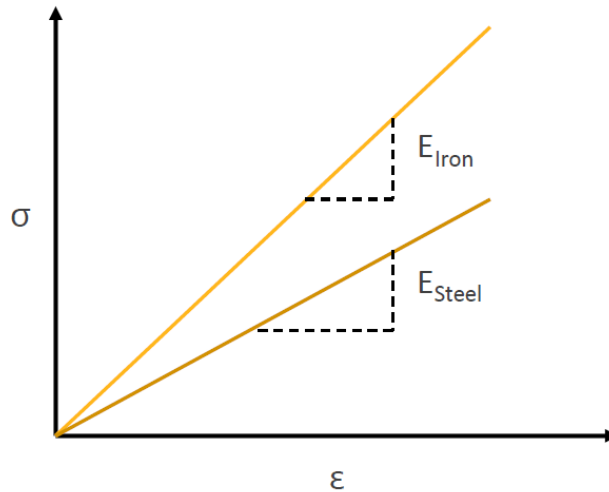


Constitutive Model

- A constitutive model is a mathematical relation between stress and strain.

$$\sigma = f(\varepsilon)$$

- The coefficients of the function are called material properties.
- Two different materials may have the same function form, but they'll have different material properties.



Cast iron is stronger...
clear choice!



Constitutive Model

- The linear elastic model is the simplest form of constitutive model.
- It's used to represent most metals.

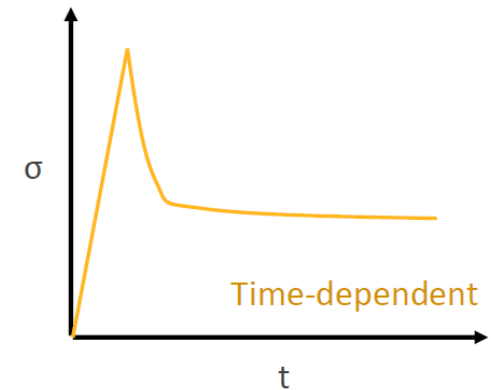
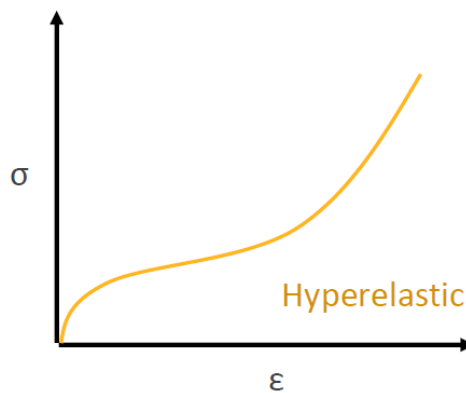
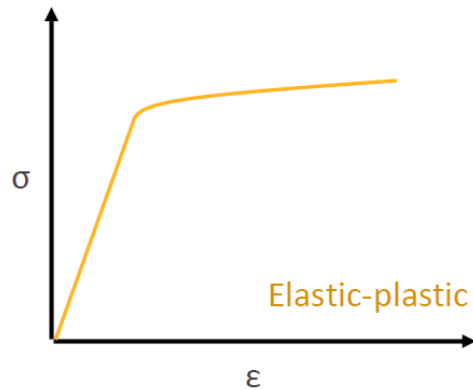
$$\sigma = E\varepsilon$$

- It may be limiting but it is very useful for most common engineering applications.
- It is governed by Hooke's law and it will be discussed in later sections.



Constitutive Model

- There are many classes of nonlinear constitutive models available to model more complex behavior.



- The choice of the material model should also depend on the physical phenomenon that we are trying to capture.
- We will discuss more in future modules.



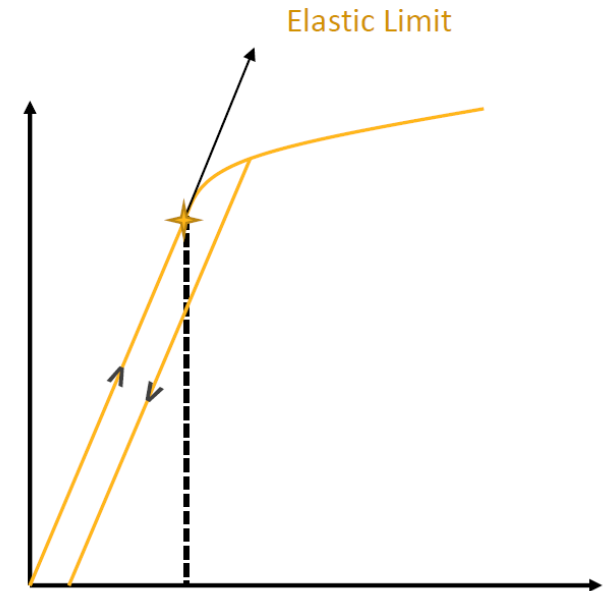
Elastic Behavior

- Elastic deformation – no permanent changes are developed in the microstructure.
- All work done on the system is stored as internal energy and is dissipated as heat energy upon unloading.
- Upon unloading, the structure returns to its original undeformed state.



Hooke's Law

- Hooke's Law – simplest form of elastic deformation.
- Most metals undergo elastic deformation up to a limit in strain – the elastic limit.
- When metals are loaded beyond this point, they sustain permanent deformation due to changes made to their microstructure.
- So, energy is not conserved beyond this point.
- Loading and unloading paths in the curve are not the same.
- But the unloading curve is parallel to the loading curve.
- Hooke's law is applicable only up to the elastic limit.

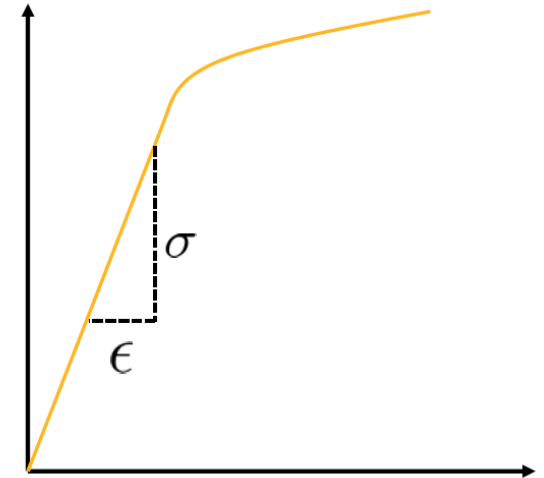


Hooke's Law

- Hooke's law states that within the elastic limit, the stress developed in the material is linearly proportional to its strain.

$$\sigma = E\epsilon$$

- The constant of proportionality is the elastic modulus also known as Young's modulus.
- In this case, the linear equation form is called the constitutive model and the Young's modulus is the material constant.
- Hooke's law is applicable to 3D cases as well.



Deformation and Strain

Strain describes the deformation of an object in a way that excludes rigid body motion.

$$\text{Deformation} = \text{Distortion} + \text{Rigid-motion}$$

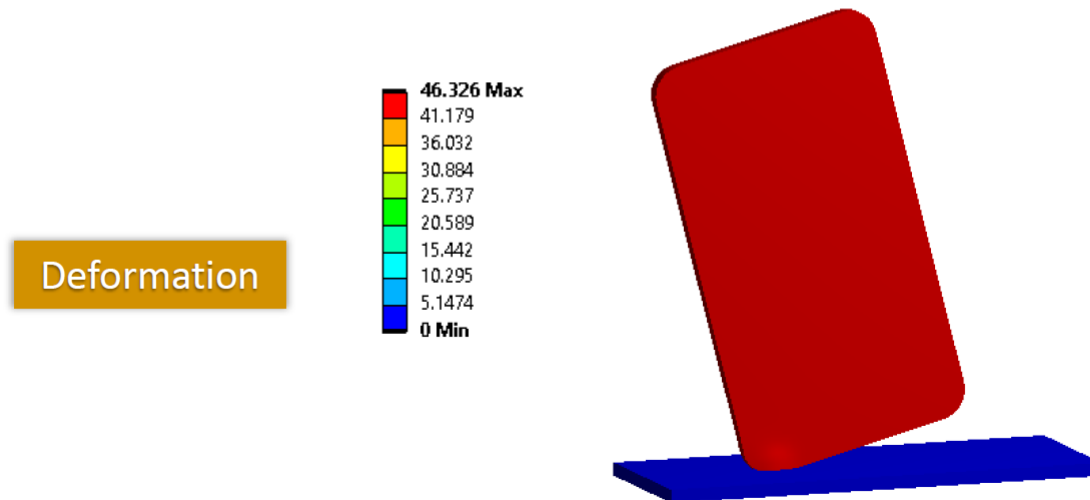
STRAIN

💡 Where does the most critical deformation occur in this cellphone drop test and how can it be quantified? The answers to these questions, in most cases, are exactly what engineers are looking for.



Deformation and Strain

If we plot the **deformation** contour of the cellphone for the dropping process, we find that the entire cellphone has very similar deformation values and the contour plot does not contain meaningful information.



Deformation and Strain

If we plot the **strain** results for the dropped cellphone, most of the strain is concentrated on the left corner. The maximum value occurs where the phone contacts the ground. The strain value is almost 0 elsewhere. Clearly the strain contour provides a more interesting presentation of the deformation, which draws our attention to the corner with the large distortion.

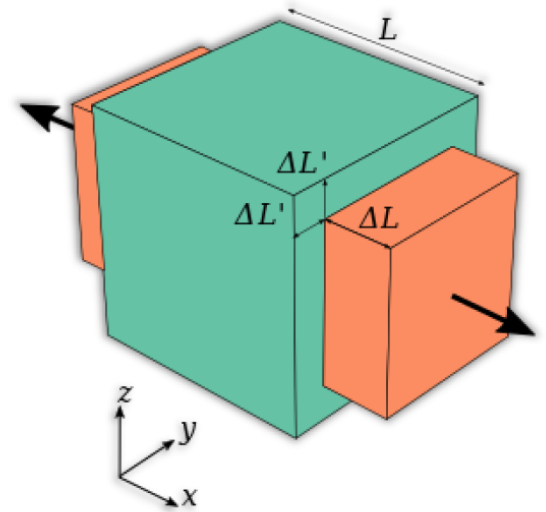


Poisson's Ratio

- Poisson's effect: when a material is stretched/compressed in one direction, it deforms in the opposite direction in other perpendicular directions.
- Poisson's ratio is defined as

$$\nu = -\frac{\epsilon_{YY}}{\epsilon_{XX}}$$

- The negative sign ensures that the value of Poisson's ratio is positive.
- Typically, the value ranges from 0 to 0.5.



Poisson's Ratio

- When a body sees deformation in one direction, it experiences strain in the other perpendicular directions.

$$\epsilon_{YY} = \epsilon_{ZZ} = -\nu \epsilon_{XX}$$

- This relation will be used in formulating strain equations using Hooke's law.



Engineering Shear Strains

- Let's review shear strain in a cube that shears in the XY plane.

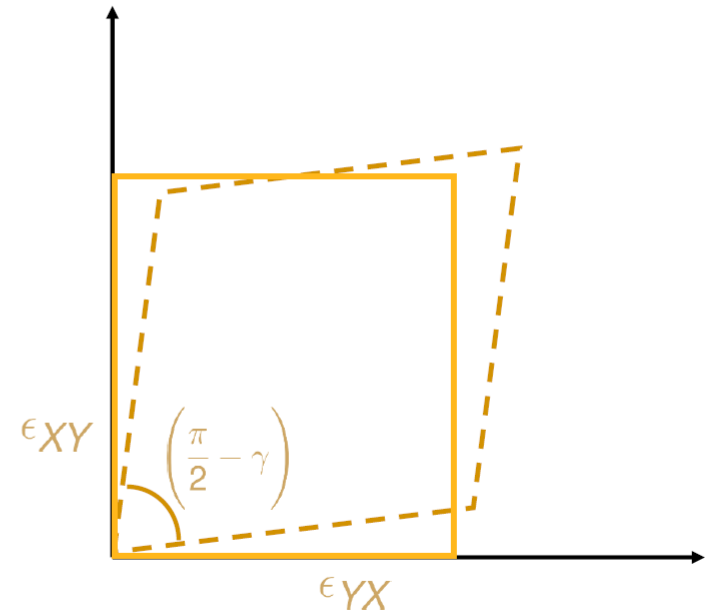
$$\gamma = \epsilon_{XY} + \epsilon_{YX}$$

- Due to symmetry,

$$\epsilon_{XY} = \epsilon_{YX}$$

- Engineering shear strain (γ) is often reported as shear strain by most commercial FE solvers.

$$\gamma = 2\epsilon_{XY} = 2\epsilon_{YX}$$



Hooke's Law 3D

- A more general case such as brake pad assembly.
- Brake pads experience both normal and shear stresses.
- All the strain components:

$$\begin{aligned}\epsilon_{XX} &= \frac{1}{E}[\sigma_{XX} - \nu(\sigma_{YY} + \sigma_{ZZ})] & \epsilon_{XY} &= \frac{(1 + \nu)}{E}\sigma_{XY} \\ \epsilon_{YY} &= \frac{1}{E}[\sigma_{YY} - \nu(\sigma_{XX} + \sigma_{ZZ})] & \epsilon_{YZ} &= \frac{(1 + \nu)}{E}\sigma_{YZ} \\ \epsilon_{ZZ} &= \frac{1}{E}[\sigma_{ZZ} - \nu(\sigma_{XX} + \sigma_{YY})] & \epsilon_{XZ} &= \frac{(1 + \nu)}{E}\sigma_{XZ}\end{aligned}$$



Strain Vector in Tensor Form

- In a full 3D case the strain tensor can be written in this form.

$$\begin{bmatrix} \epsilon_{XX} \\ \epsilon_{YY} \\ \epsilon_{ZZ} \\ 2\epsilon_{YZ} \\ 2\epsilon_{XZ} \\ 2\epsilon_{XY} \end{bmatrix} = \frac{1}{E} \begin{bmatrix} 1 & -\nu & -\nu & 0 & 0 & 0 \\ -\nu & 1 & -\nu & 0 & 0 & 0 \\ -\nu & -\nu & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2(1+\nu) & 0 & 0 \\ 0 & 0 & 0 & 0 & 2(1+\nu) & 0 \\ 0 & 0 & 0 & 0 & 0 & 2(1+\nu) \end{bmatrix} \begin{bmatrix} \sigma_{XX} \\ \sigma_{YY} \\ \sigma_{ZZ} \\ \sigma_{YZ} \\ \sigma_{XZ} \\ \sigma_{XY} \end{bmatrix}$$

- It is in full agreement with the linear form proposed by Hooke's law.
- The constant tensor is called the Compliance tensor.

$$\epsilon = \mathbf{C}\sigma$$



Stress Vector in Tensor Form

- Due to its linear nature, the same equation can be expressed in the form of stress vector.

$$\begin{bmatrix} \sigma_{XX} \\ \sigma_{YY} \\ \sigma_{ZZ} \\ \sigma_{YZ} \\ \sigma_{XZ} \\ \sigma_{XY} \end{bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & \nu & 0 & 0 & 0 \\ \nu & 1-\nu & \nu & 0 & 0 & 0 \\ \nu & \nu & 1-\nu & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{(1-2\nu)}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{(1-2\nu)}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{(1-2\nu)}{2} \end{bmatrix} \begin{bmatrix} \epsilon_{XX} \\ \epsilon_{YY} \\ \epsilon_{ZZ} \\ \epsilon_{YZ} \\ \epsilon_{XZ} \\ \epsilon_{XY} \end{bmatrix}$$

- It still follows the Hooke's law and the tensor of constants is called the Stiffness matrix.
- This is what Hooke's law looks like in a 3D isotropic body.

$$\sigma = \mathbf{E}\epsilon$$

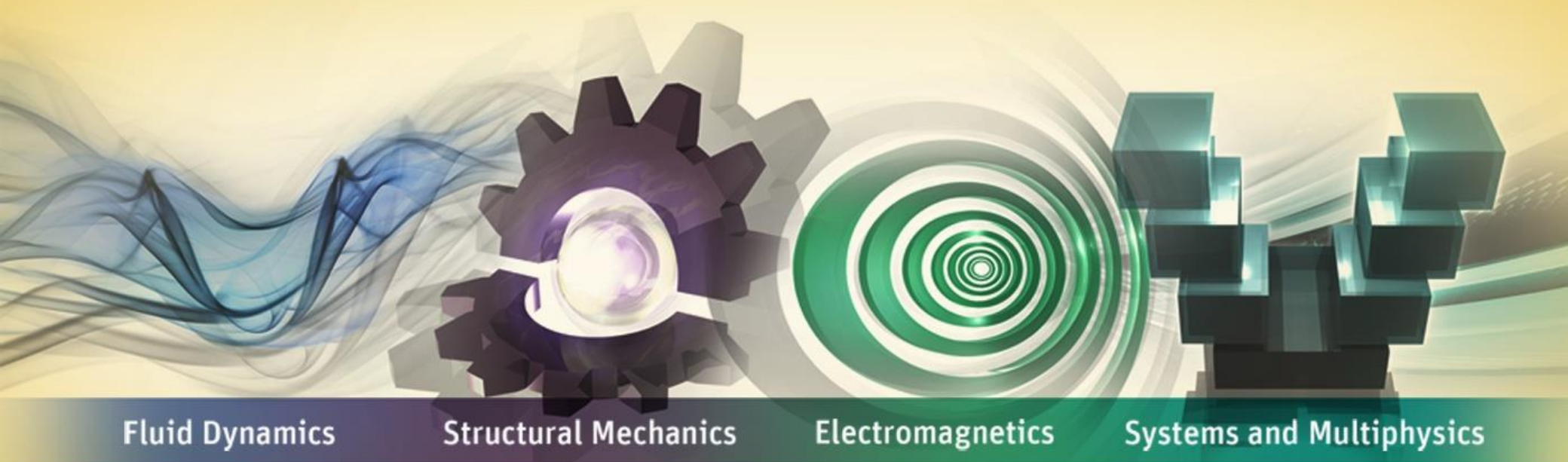


Anisotropic Materials

- In reality, not all materials are isotropic; some have significant directional behavior.
- In such cases the Young's modulus and Poisson's ratio change according to the principal direction.
- Examples include various materials such as wood, fibrous tissues, composites, rolled metal specimens, etc.
- Hooke's law is still applicable, but the equations need to account for the directional material properties.



Lecture 1: Overview



ANSYS Mechanical

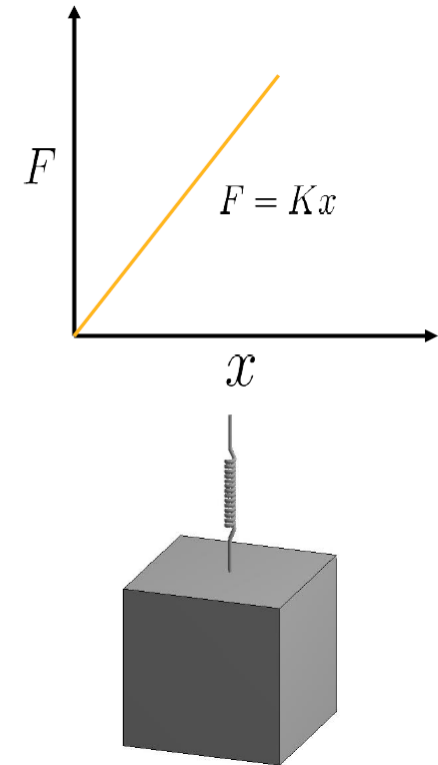
1.2 Introduction to Structural Nonlinearities

Why Study Nonlinearities?

- Mechanical systems whose force-displacement response is linear are called linear systems.
- A simple spring-weight is an example of a linear system.
- Spring stiffness remains constant throughout deformation, which is why the force response for a given displacement increases linearly:

$$F = Kx$$

- In reality, mechanical systems are not linear but some of them may be idealized as linear to simplify analysis.

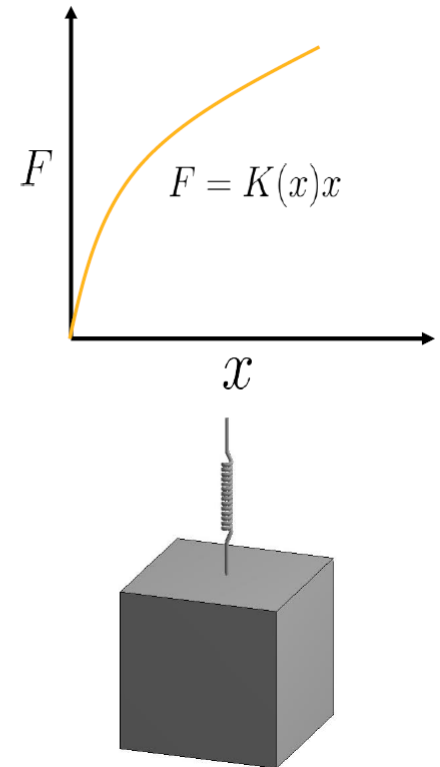


Why Study Nonlinearities?

- In reality, the stiffness of a system is not constant and changes with deformation.
- Because of this, the force-displacement relation takes a nonlinear form:

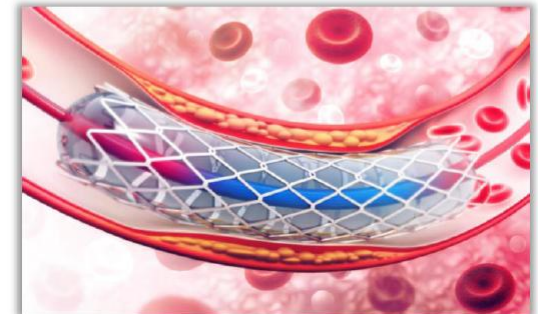
$$F = K(x)x$$

- Depending on how $K(x)$ changes, the complexity of solving the equation changes.
- But why can't we simplify this into a linear system?



Importance of Nonlinearity

- In most real life and industrial applications, nonlinearities are too important to be neglected.
- Very large deformations witnessed during crashworthiness tests are not captured using linear analysis.
- Head impact simulations are inaccurate when material nonlinearities are neglected.
- Stent deployment forces are over-predicted when linearity is assumed, which can be fatal for patients.
- So what introduces nonlinearities?



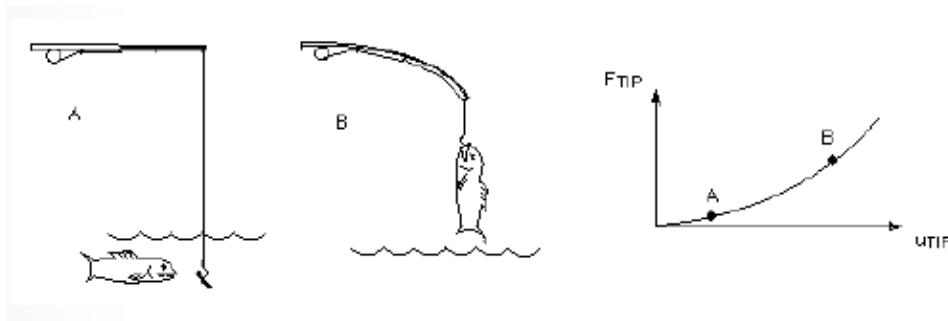
Types of Nonlinearities

- There are three factors that introduce nonlinearities into mechanical systems:
 1. Large deformations
 2. Material nonlinearity
 3. Contacts
- Each of these factors results in a change of stiffness in the system with deformation.
- Often, real applications have more than one source of nonlinearity.
- As the complexity of the system increases, linear solvers are no longer sufficient for solving such systems of equations.



Large Deformations

- Deformations are considered large when there's a noticeable change in the shape of the structure.
- Change in shape can result in
 1. Change of wall thickness
 2. Change of area of cross-section
 3. Change in direction of loading
- All of the above result in nonlinear responses.

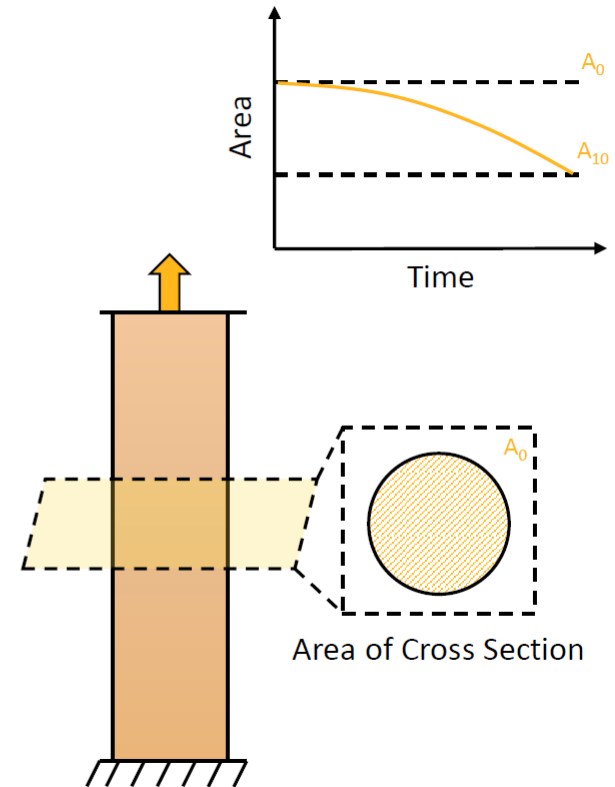


- Changing geometric configuration can cause nonlinear behavior



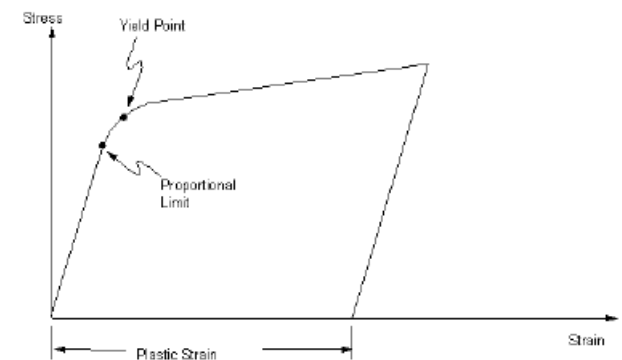
Large Deformations

- Consider a metal coupon under tensile loading.
- The initial force-displacement response recorded may be approximated to be linear.
- At larger deformation, the cross-section of the coupon decreases and results in a nonlinear response.
- This clearly shows how changes in shape at large deformation can introduce nonlinearity.



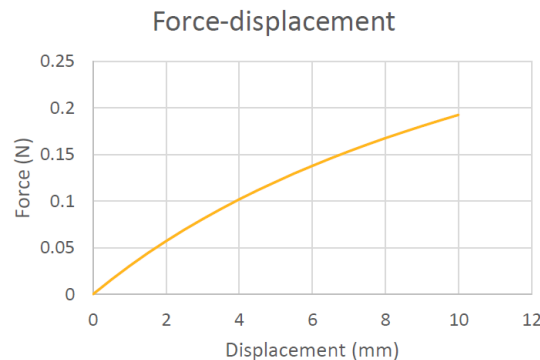
Material Nonlinearity

- All materials have nonlinear behavior to some degree.
- Depending on the application and the amount of deformation, some materials may be assumed to be linear in nature.
- Some materials or applications, however, cannot be assumed to be linear.
- Consider rolling of metals in which a rigid roller is used to deform a metal piece into a desired shape or thickness.
- Metal plasticity is the key material behavior to capture this process.
- Ignoring plasticity yields incorrect results.



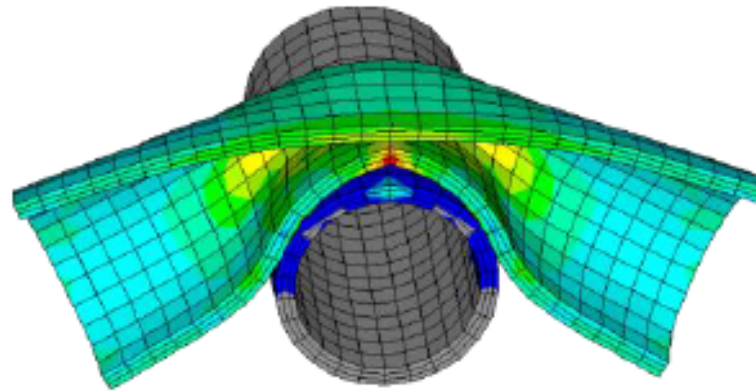
Material Nonlinearity

- In this example we demonstrate how material nonlinearity introduces nonlinearity into the total response.
- A dog-bone sample is attached to a tensile testing instrument. One end is fixed while the other end is stretched; the reaction force is recorded by the load cell.
- In case of neoprene rubber, the force displacement response is no longer linear.
- This is because a nonlinear elastic behavior is used for defining the material behavior.
- This shows that using a nonlinear material changes the nature of the system to nonlinear.



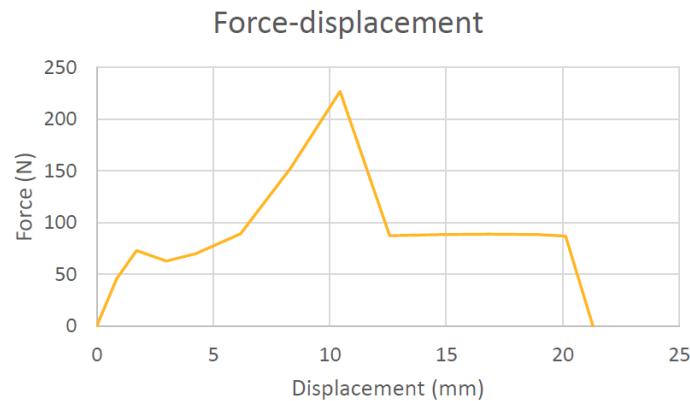
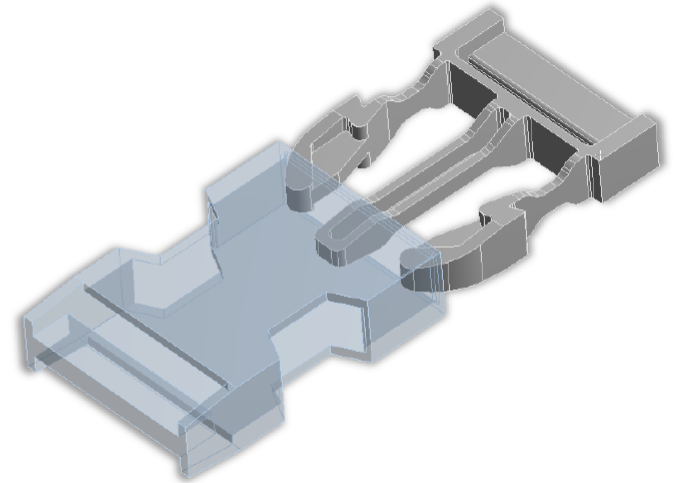
Contacts

- Contact is a “changing status” nonlinearity, where an abrupt change in stiffness may occur when bodies come into or out of contact with each other.



Contacts

- In this example we will discuss a special case in which a loss of contact introduces nonlinearity.
- A very common occurrence of this is seen in a snap-fit buckle.
- When the male part settles inside the female component, there is a sudden loss of stiffness due to loss of contact.

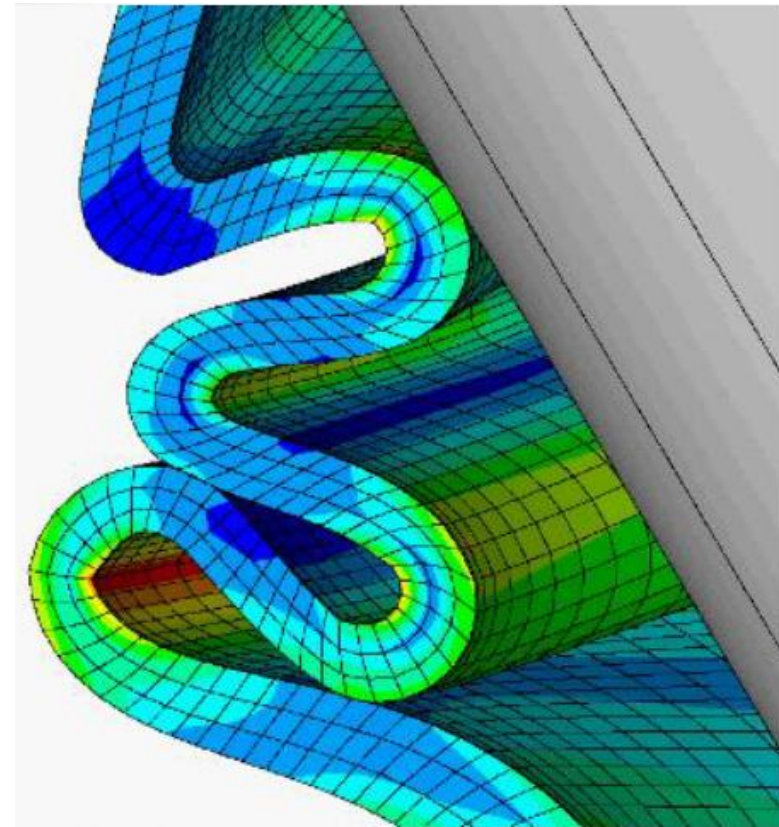


- Due to the sudden change in stiffness, the force response drops too.
- This introduces nonlinearity in the model response.



Combination of Nonlinearities

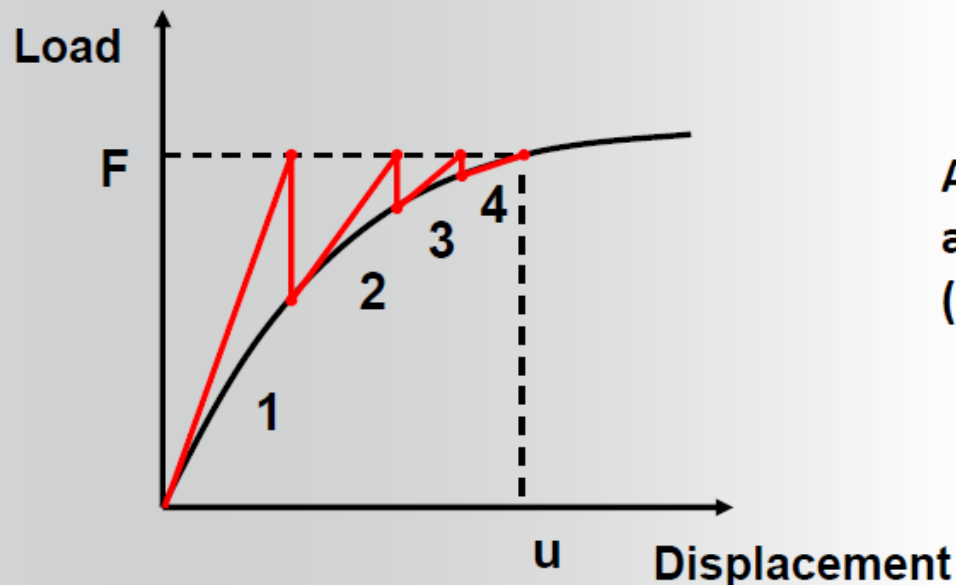
An example of nonlinear geometry (large strain and large deformation), nonlinear material (rubber), and changing status nonlinearities (contact)



B. Nonlinear solution using linear solvers

How does *Mechanical* solve for a changing stiffness?

- In a nonlinear analysis, the response cannot be predicted directly with a set of linear equations.
- However, a nonlinear structure can be analyzed using an iterative series of linear approximations, with corrections.
- *Mechanical* uses an iterative process called the *Newton-Raphson Method*. Each iteration is known as an *equilibrium iteration*.



A full Newton-Raphson iterative analysis for one increment of load. (Four iterations are shown.)

Newton-Raphson Method

To solve nonlinear equations, Newton-Raphson method is usually used. To see how it works, consider a single variable scenario.

If the function is continuous, Taylor series expansion can be performed:

$$f(x^{(i+1)}) \approx f(x^{(i)}) + \frac{\partial f(x^{(i)})}{\partial x} (x^{(i+1)} - x^{(i)}) + \frac{1}{2!} \left(\frac{\partial^2 f(x^{(i)})}{\partial x^2} \right) (x^{(i+1)} - x^{(i)})^2 + \dots$$

Neglecting high-order terms:

$$f(x^{(i+1)}) \approx f(x^{(i)}) + \frac{\partial f(x^{(i)})}{\partial x} (x^{(i+1)} - x^{(i)})$$

Rearranging:

$$(x^{(i+1)} - x^{(i)}) = \frac{f(x^{(i+1)}) - f(x^{(i)})}{f'(x^{(i)})}$$

$$\Delta x = \frac{F - f(x^{(i)})}{K_t} = K_t^{-1} (F - f(x^{(i)}))$$

$$\Delta x = (x^{(i+1)} - x^{(i)})$$

$$K_t = f'(x^{(i)})$$



Newton-Raphson Method

$$\Delta x = K_t^{-1} \left(F - f(x^{(i)}) \right)$$

F = desired value

$x^{(i)}$ = initial value of x

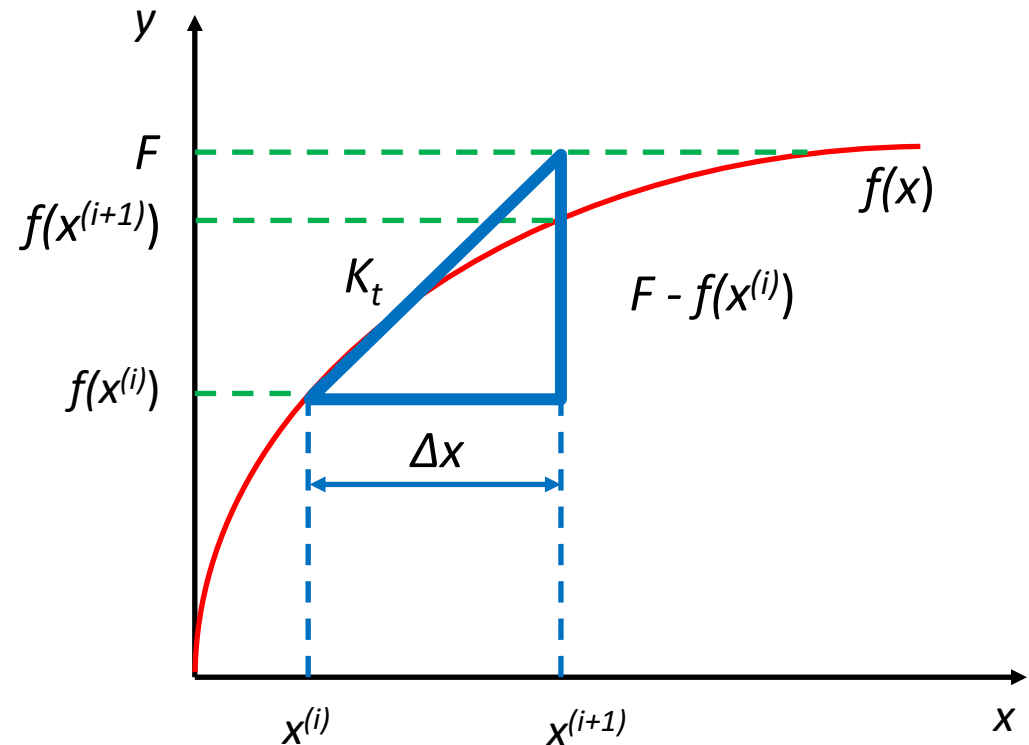
$f(x^{(i)})$ = initial value of the function

K_t = slope of the tangent line

Δx = increment in x

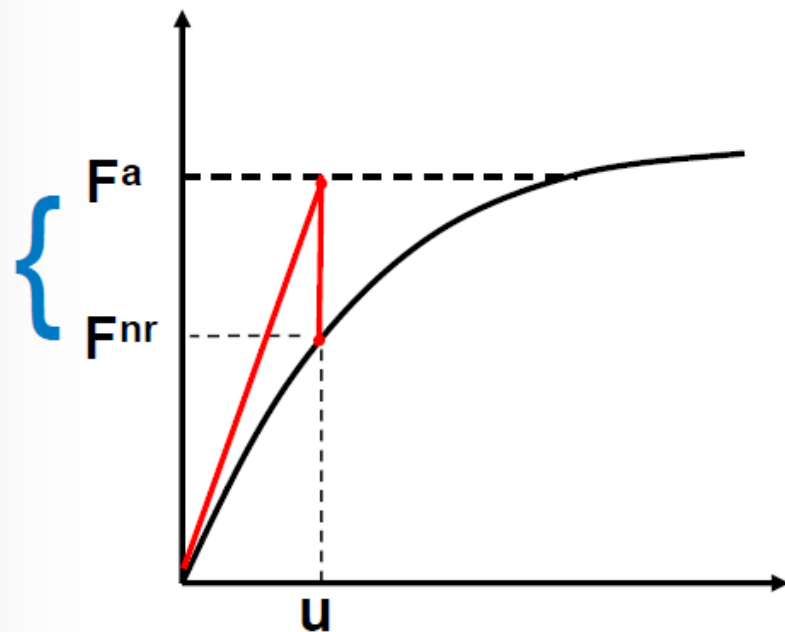
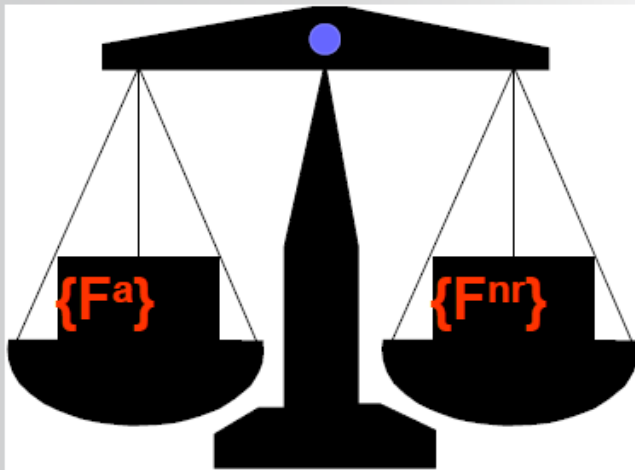
In FEA:

- $F - f(x^{(i)})$ = difference between applied force (F) and internal forces $f(x^{(i)})$
- Δx = displacement increment



... Nonlinear solution using linear solvers

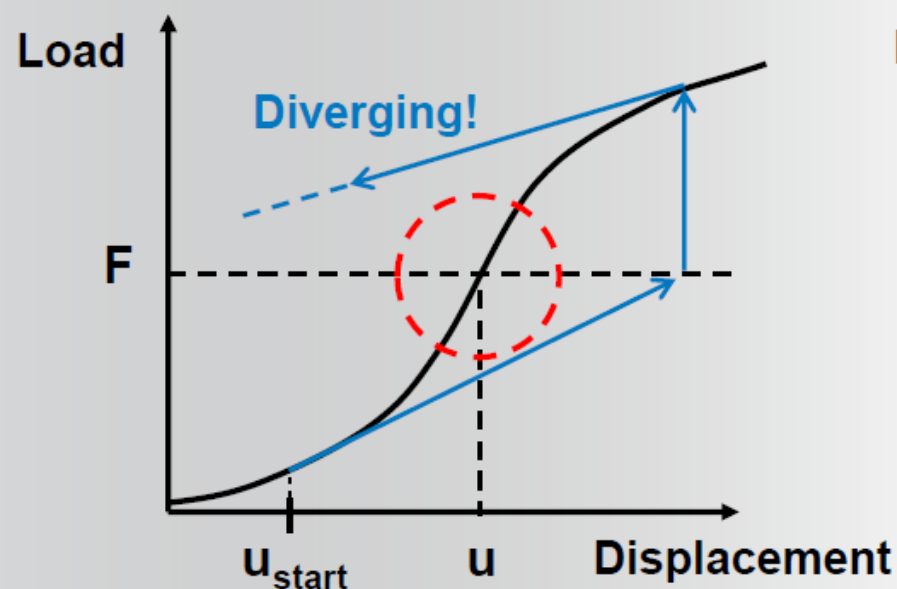
- The difference between external and internal loads, $\{F^a\} - \{F^{nr}\}$, is called the *residual*. It is a measure of the force imbalance in the structure.
- The goal is to iterate until the residual becomes acceptably small; that is, until the solution is *converged*.
- When convergence is achieved, the solution is in *equilibrium*, within an acceptable tolerance.



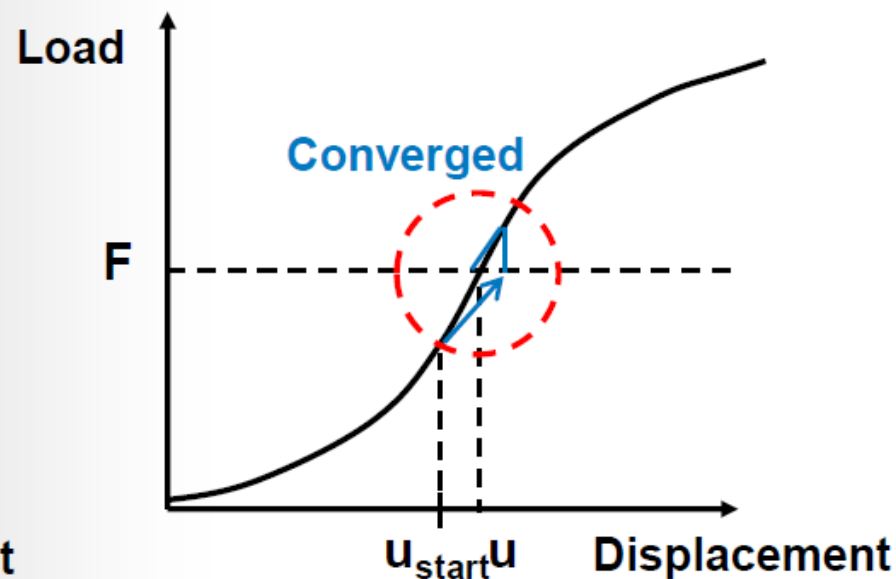
... Nonlinear solution using linear solvers

The Newton-Raphson method:

- Is *not* guaranteed to converge in all cases!
- Will converge only if the starting configuration is inside the *radius of convergence*.



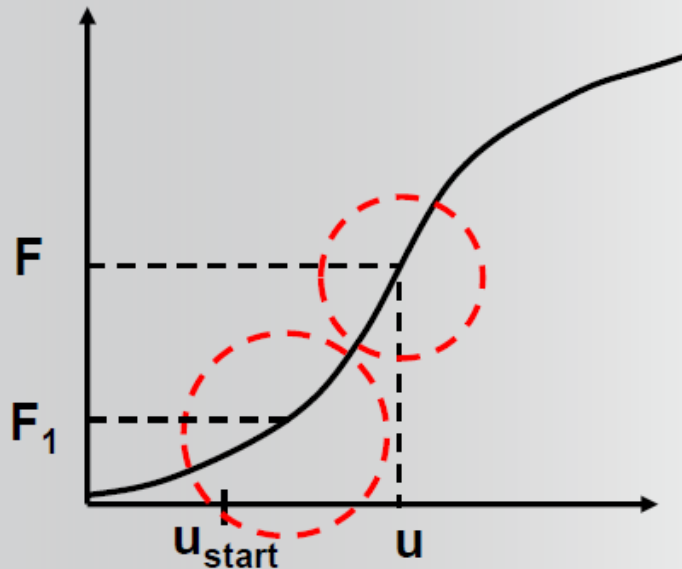
Starting outside the
radius of convergence



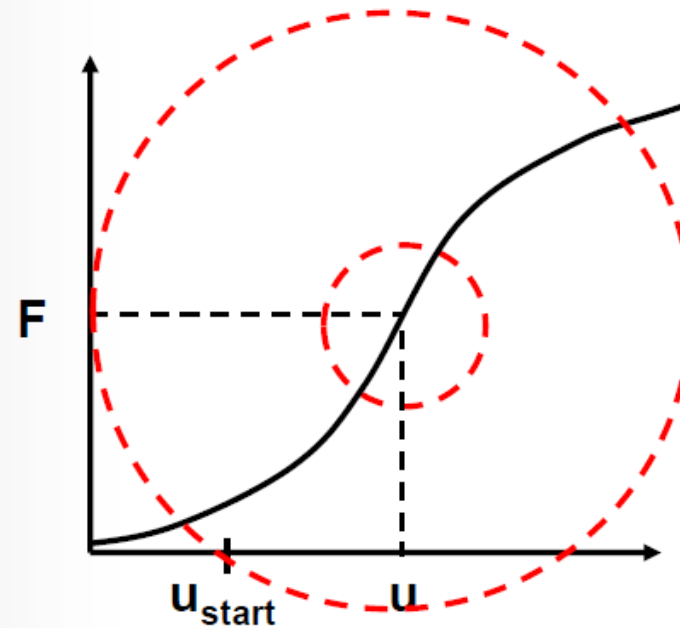
Starting inside the radius
of convergence

... Nonlinear solution using linear solvers

Two techniques can help you obtain a converged solution:



Apply load incrementally to move the target closer to the start



Use convergence-enhancement tools to enlarge the radius of convergence

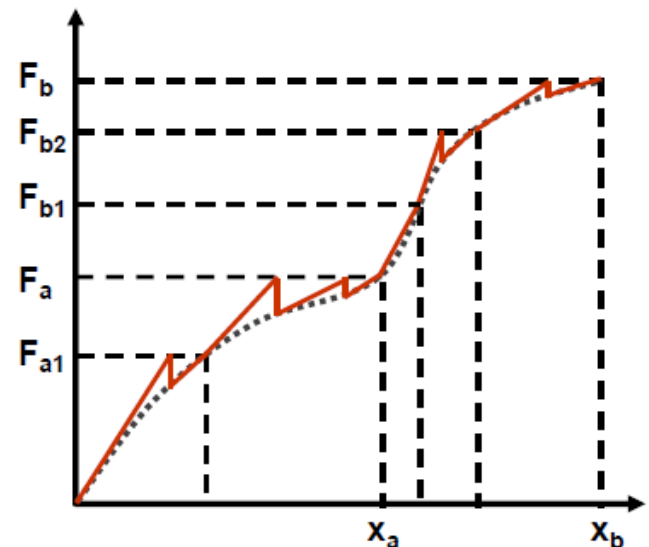
Mechanical combines both strategies to obtain convergence.

... Nonlinear solution using linear solvers

As a general rule, sudden changes to any aspect of a system will cause convergence difficulties.

With this in mind, it is useful to understand how loads are managed

- **Load steps** differentiate changes in general loading.
 - In the Figure at the bottom right, F_a and F_b are loadsteps.
- **Substeps** apply the loads in an incremental fashion
 - Because of the complex response, it may be necessary to apply the load incrementally. For example, F_{a1} may be near 50% of the F_a load. After the load for F_{a1} is converged, then the full F_a load is applied. F_a has 2 substeps while F_b has 3 substeps in this example
- **Equilibrium iterations** are the corrective solutions to obtain a converged substep
 - In this example, the iterations between the dotted lines indicate equilibrium iterations.



C. Nonlinear FEA Issues

Three main issues arise whenever you do a nonlinear finite element analysis:

- Obtaining convergence
- Balancing expense versus accuracy
- Verification

It takes care and skill to
juggle these three
issues successfully!



Obtaining convergence...

Usually your biggest challenge.

Solution must start within the radius of convergence.

- The radius of convergence is unknown!
 - If solution converges, the start was within the radius.
 - If solution fails to converge, the start was outside the radius.
- Trial-and-error is sometimes required.
- Experience and training reduce your trial-and-error effort.

Difficult problems might require many load increments, and many iterations at each load increment, to reach convergence.

- When many iterations are required, the overall solution time increases.



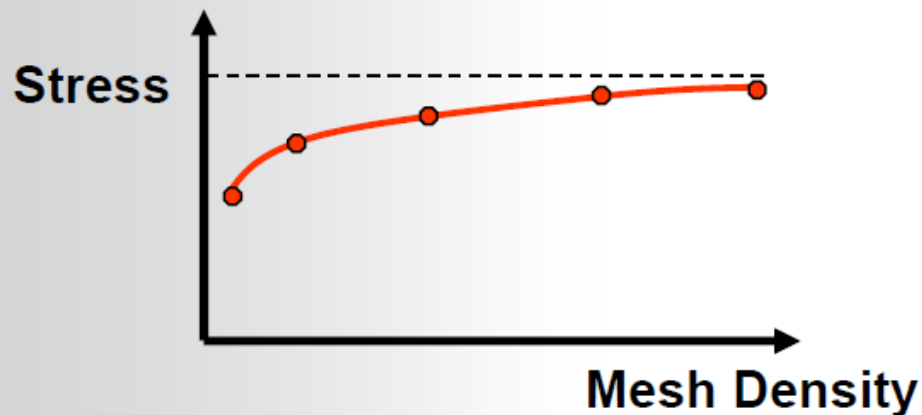
Balancing expense versus accuracy...

- All FEA involves a trade-off between expense (elapsed time, disk and memory requirements) and accuracy.
- More detail and a finer mesh generally lead to a more accurate solution, but require more time and system resources.
- Nonlinear analyses add an extra factor, *the number of load increments*, which affects both accuracy and expense. More increments = improve the accuracy, with increase the expense.
- Other nonlinear parameters, such as contact stiffness (discussed later), can also affect both accuracy and expense.
- Use your own engineering judgment to determine how much accuracy you need, how much expense you can afford.



Verification...

- In a nonlinear analysis, as in any finite-element analysis, you must *verify* your results.
- Due to the increased complexity of nonlinear behavior, nonlinear results are generally more difficult to verify.
- Sensitivity studies (increasing mesh density, decreasing load increment, varying other model parameters) become more expensive.



Typical Sensitivity Study

Later chapters will provide modeling tips for different nonlinear situations.